

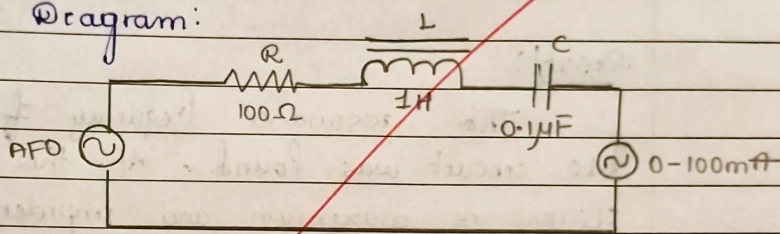
* Exp No-1 *

* ANALYSIS OF SERIES RESONANCE CONCEPT *

aim:

To practically observe the behaviour of a series resonance circuit and plot the performance curves. current v/s frequency, Impedance v/s frequency, X_L v/s frequency, X_C v/s Frequency, Phase angle v/s frequency.

App. Circuit Diagram:



Sl. No	Freq. m (Hz)	Ammeter reading (mA)	$X_L = 2\pi fL$ (m ohms)	$X_C = 1/2\pi fC$ (m ohms)	Z (m ohms)	θ m degree
1	50	0.04	314.16	31830.99	31516.99	-89.82°
2	80	0.09	502.65	19894.37	19391.98	-89.7°
3	100	0.13	628.32	15915.49	15287.5	-89.63°
4	150	0.22	942.48	10610.33	9668.37	-89.4°
5	200	0.33	1256.64	7957.75	6701.86	-89.14°
6	250	0.49	1570.8	6366.2	4796.44	-89.05°
7	300	0.73	1884.96	5305.16	3421.66	-88.32°
8	350	1.14	2199.11	4547.28	2350.3	-87.6°
9	400	1.97	2513.27	3978.87	1469.00	-86.1°
10	450	3.97	2827.43	3536.78	716.36	-82.0°
11	500	5.14	3141.6	3183.1	99.8	-22.5°
12	550	1.77	3455.75	2893.73	570.85	+80.0°
13	600	1.08	3769.91	2652.6	1121.78	85.0°
14	650	0.83	4084.07	2448.54	1638.58	86.5°
15	700	0.66	4398.23	2273.64	2126.94	87.3°
16	750	0.56	4712.39	2122.06	2592.26	87.8°
17	800	0.48	5026.55	1989.44	3038.75	88.1°

calculation:

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$\theta = \tan^{-1} (X_L - X_C) / R$$

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{1 \times 0.1 \times 10^{-6}}} = \underline{\underline{503.3 \text{ Hz}}}$$

Result & Inference:

Result:

The resonance frequency of the series RLC circuit was found. At this frequency, current is maximum and impedance is minimum.

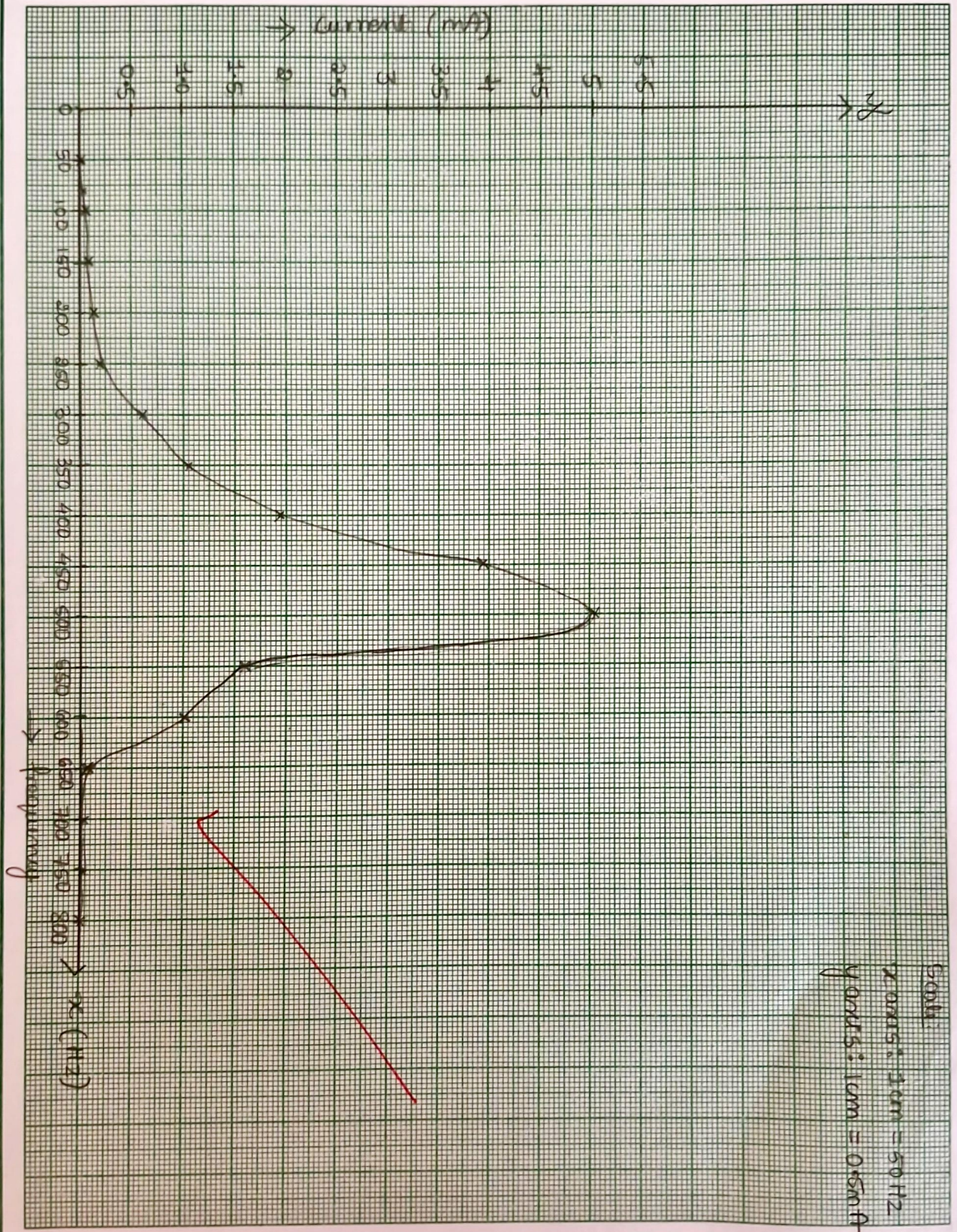
Inference:

At resonance, the circuit acts like a pure resistor. Below resonance it is capacitive, and above resonance it is inductive.

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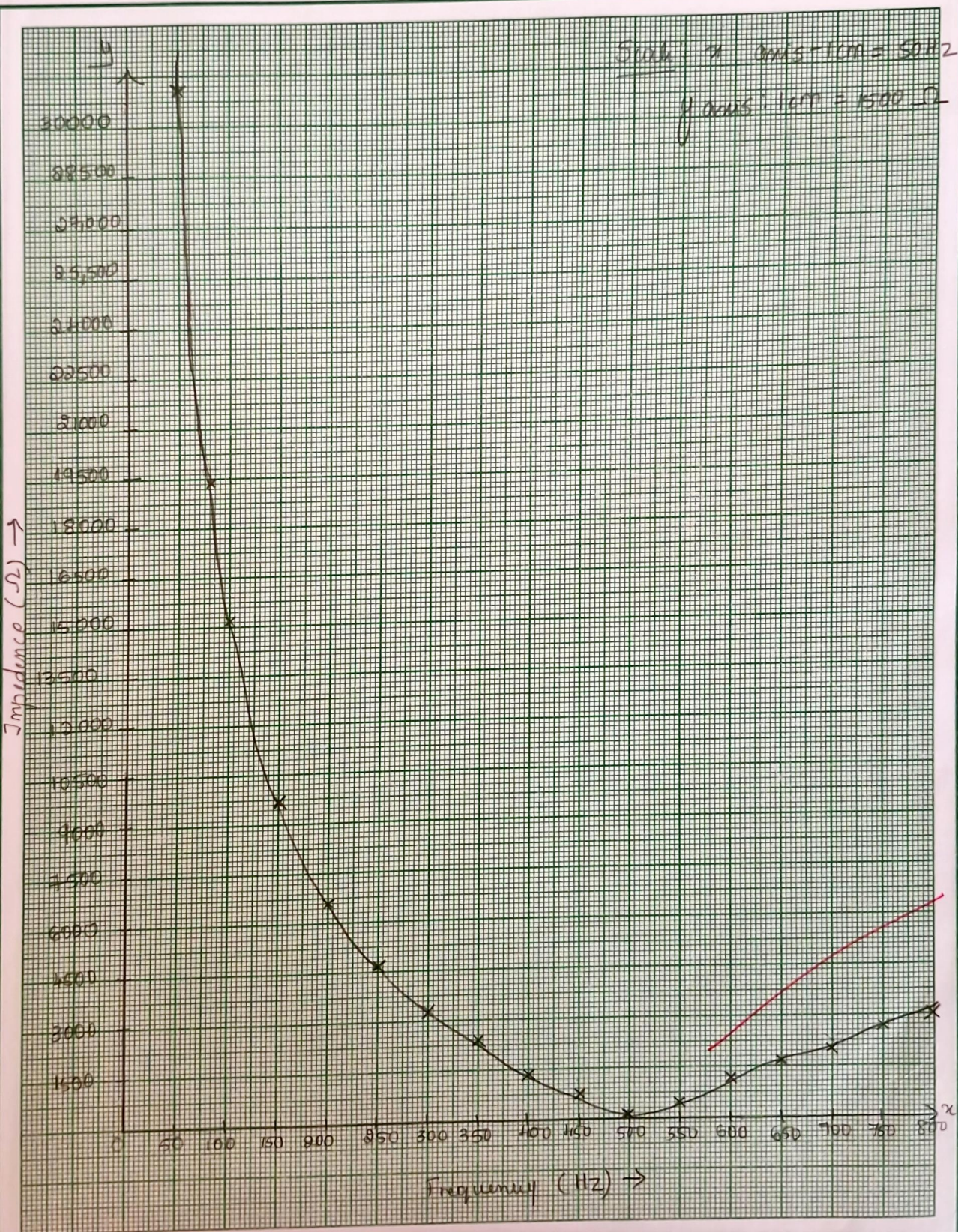
1] Resonance Curve (Current vs Frequency)

Date: 14/08/2025



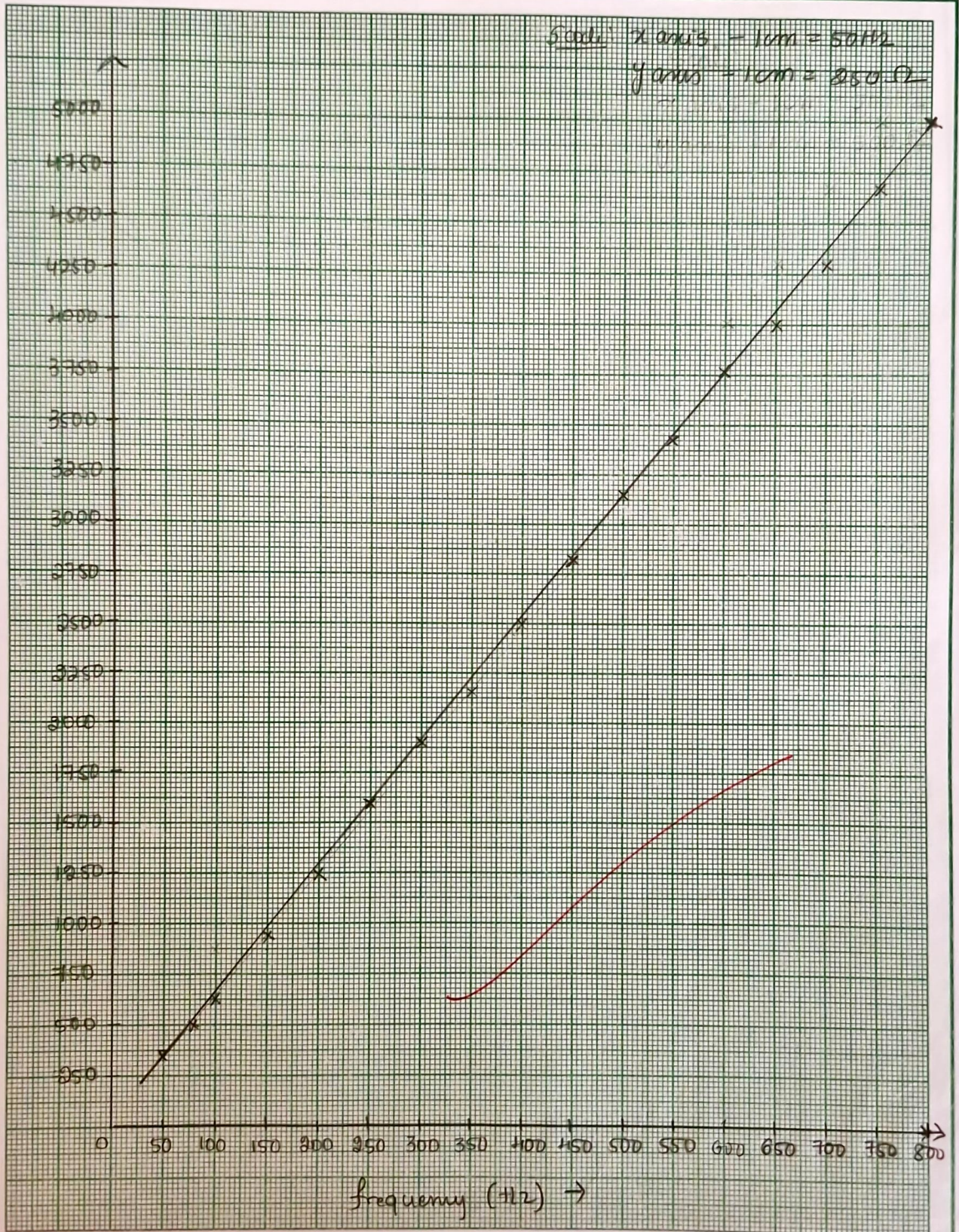
i) Impedance vs Frequency

Date: 14/03/2025



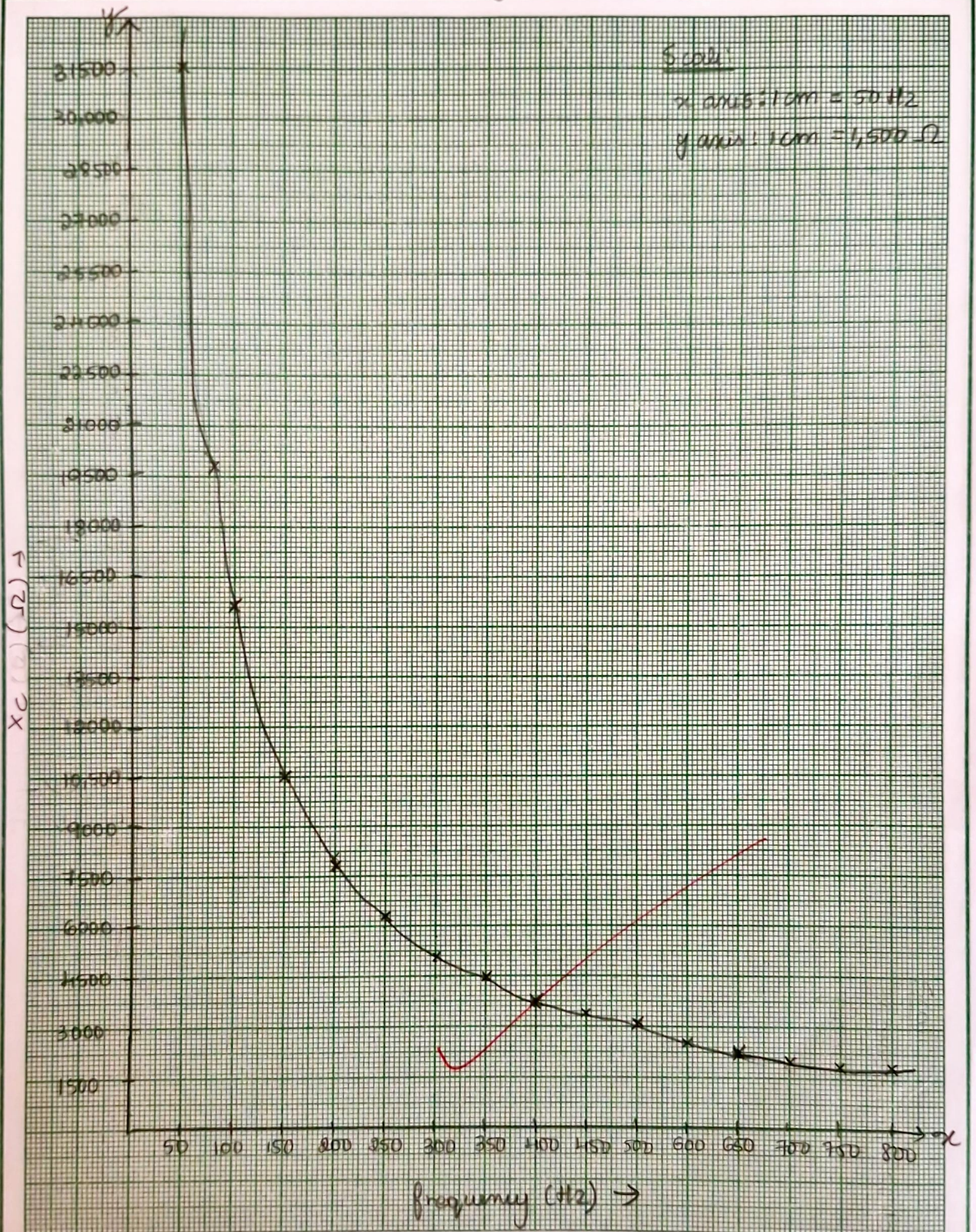
1117] X_L vs frequency.

Date : 14/08/2025



iv) X_C vs frequency

Date: 14/08/2025



v) Phase angle vs frequency.

Date: 14/08/2025

Scale: $\pi \text{ cm} \times 1 \text{ cm} = 10 \text{ Hz}$
 $1 \text{ cm} \times 1 \text{ cm} = 10^\circ$

